

CSI132 — Discrete Structures II

Tutorial on Functions — Complete Solutions

Duration: 2 hours | All Questions Attempted
Complete solutions with visual demonstrations and detailed explanations

Question 1: Determining if Relations are Functions

- Determine whether f is a function from $\mathbb{Z}$ to $\mathbb{R}$:

(a) $f(n) = \pm n$

Answer: **NOT a function** ✗

Explanation: For each input n , there are two possible outputs: $+n$ and $-n$. A function must assign exactly one output to each input. For example, when $n = 5$, we get both $f(5) = 5$ and $f(5) = -5$, which violates the definition of a function.

Input (n)	Possible Output 1	Possible Output 2
5	+5	-5
3	+3	-3
-2	$+(-2) = -2$	$-(-2) = +2$

(b) $f(n) = \sqrt{n^2 + 1}$

Answer: **YES, this is a function** ✓

Explanation: For every integer $n \in \mathbb{Z}$, the expression $n^2 + 1$ is always positive (since $n^2 \geq 0$), and the square root of a positive number is uniquely defined as the positive square root. Thus, each n maps to exactly one real number.

(c) $f(n) = 1/(n^2 - 4)$

Answer: **NOT a function** ✗

Explanation: The expression is undefined when $n^2 - 4 = 0$, which occurs at $n = 2$ and $n = -2$. Since these values are in the domain $\mathbb{Z}$ but have no corresponding output, this is not a function from $\mathbb{Z}$ to $\mathbb{R}$.

(d) $f(n) = 1/n$

Answer: **NOT a function** ✗

Explanation: The expression is undefined when $n = 0$. Since $0 \in \mathbb{Z}$ but $f(0)$ is undefined, this is not a function from $\mathbb{Z}$ to $\mathbb{R}$.

Question 2: Domain and Range

(a) Function assigning first integer of pair

Domain: $\mathbb{Z}^+ \times \mathbb{Z}^+ = \{(m,n) : m, n \in \mathbb{Z}^+\}$

The set of all ordered pairs of positive integers

Range: $\mathbb{Z}^+ = \{1, 2, 3, 4, \dots\}$

The set of all positive integers

Justification: For any positive integer k , we can find a pair $(k, 1)$ that maps to k .

(b) Function assigning (ones – zeros) to bit strings

Domain: $\{0, 1\}^*$

The set of all bit strings

Range: $\mathbb{Z}$ (all integers)

Explanation: For a bit string of length n with k ones, it has $(n-k)$ zeros, so the function value is $k - (n-k) = 2k - n$. We can achieve any integer: for $m \geq 0$, use m ones and 0 zeros; for $m < 0$, use 0 ones and $|m|$ zeros.

Examples:

Bit String	Ones – Zeros	Result
111	3 – 0	+3
0000	0 – 4	–4
101	2 – 1	+1
00	0 – 2	–2

Question 3: Floor and Ceiling Functions

Part	Expression	Answer	Explanation
(a)	$\lfloor 1.1 \rfloor$	1	Greatest integer ≤ 1.1
(b)	$\lceil 1.1 \rceil$	2	Smallest integer ≥ 1.1
(c)	$\lfloor -0.1 \rfloor$	-1	Greatest integer ≤ -0.1
(d)	$\lceil -0.1 \rceil$	0	Smallest integer ≥ -0.1
(e)	$\lceil 2.99 \rceil$	3	Smallest integer ≥ 2.99
(f)	$\lfloor -2.99 \rfloor$	-2	Smallest integer ≥ -2.99
(g)	$\lfloor \frac{1}{2} + \lceil \frac{1}{2} \rceil \rfloor$	1	$\lceil \frac{1}{2} \rceil = 1$, then $\lfloor 1.5 \rfloor = 1$
(h)	$\lceil \lfloor \frac{1}{2} \rfloor + \lceil \frac{1}{2} \rceil + \frac{1}{2} \rceil$	2	$0 + 1 + 0.5 = 1.5$, $\lceil 1.5 \rceil = 2$

Key Concepts:

- Floor function $\lfloor x \rfloor$: Returns the greatest integer less than or equal to x
- Ceiling function $\lceil x \rceil$: Returns the smallest integer greater than or equal to x
- For negative numbers, floor rounds down (away from zero) and ceiling rounds up (toward zero)

Question 4: One-to-One Functions (Finite Sets)

Determine whether each function from $\{a, b, c, d\}$ to itself is one-to-one:

(a) $f(a)=b, f(b)=a, f(c)=c, f(d)=d$

Answer: **ONE-TO-ONE** ✓

Verification: All outputs are distinct. No two different inputs map to the same output.

Input	Output
a	b
b	a
c	c
d	d

(b) $f(a)=b, f(b)=b, f(c)=d, f(d)=c$

Answer: **NOT ONE-TO-ONE** ✗

Problem: Both a and b map to b. Since $f(a) = f(b) = b$ but $a \neq b$, the function is not one-to-one.

Input	Output
a	b
b	b
c	d
d	c

(c) $f(a)=d, f(b)=b, f(c)=c, f(d)=d$

Answer: **NOT ONE-TO-ONE** ✗

Problem: Both a and d map to d. Since $f(a) = f(d) = d$ but $a \neq d$, the function is not one-to-one.

Question 5: One-to-One Functions from $\mathbb{Z}$ to $\mathbb{Z}$

Part	Function	One-to-One?	Key Insight
(a)	$f(n) = n - 1$	YES ✓	Linear shift preserves uniqueness
(b)	$f(n) = n^2 + 1$	NO X	$f(-1) = f(1) = 2$
(c)	$f(n) = n^3$	YES ✓	Cubic is strictly monotonic
(d)	$f(n) = \lceil n/2 \rceil$	NO X	$f(1) = f(2) = 1$

Detailed Analysis:

(a) $f(n) = n - 1$

Proof: Suppose $f(n_1) = f(n_2)$. Then $n_1 - 1 = n_2 - 1$, which implies $n_1 = n_2$. $\therefore$ One-to-one.

(b) $f(n) = n^2 + 1$

Counterexample: $f(-1) = (-1)^2 + 1 = 2$ and $f(1) = 1^2 + 1 = 2$. Since $f(-1) = f(1)$ but $-1 \neq 1$, not one-to-one.

(c) $f(n) = n^3$

Proof: Suppose $f(n_1) = f(n_2)$. Then $n_1^3 = n_2^3$, which implies $n_1 = n_2$ (cube function is strictly monotonic). $\therefore$ One-to-one.

(d) $f(n) = \lceil n/2 \rceil$

Counterexample: $f(1) = \lceil 1/2 \rceil = \lceil 0.5 \rceil = 1$ and $f(2) = \lceil 2/2 \rceil = \lceil 1 \rceil = 1$. Since $f(1) = f(2)$ but $1 \neq 2$, not one-to-one.

Question 6: Onto Functions from $\mathbb{Z} \times \mathbb{Z}$ to $\mathbb{Z}$

Part	Function	Onto?	Key Finding
(a)	$f(m,n) = 2m - n$	YES ✓	Use $(0, -k)$ to get k
(b)	$f(m,n) = m^2 - n^2$	NO ✗	Cannot achieve 2
(c)	$f(m,n) = m + n + 1$	YES ✓	Use $(k-1, 0)$ to get k
(d)	$f(m,n) = m - n $	YES ✓	Range covers all $\mathbb{Z}$
(e)	$f(m,n) = m^2 - 4$	NO ✗	Cannot achieve 1

Detailed Proofs:

(a) $f(m,n) = 2m - n$ is **ONTO**

Proof: For any $k \in \mathbb{Z}$, choose $m = 0$ and $n = -k$. Then $f(0, -k) = 2(0) - (-k) = k$. $\therefore$ Onto.

(b) $f(m,n) = m^2 - n^2$ is **NOT ONTO**

Counterexample: Consider $k = 2$. We need $m^2 - n^2 = 2$, or $(m-n)(m+n) = 2$. For integer solutions, we need $(m-n)$ and $(m+n)$ to be divisors of 2. Testing all cases shows no integer solutions exist. $\therefore$ Not onto.

(c) $f(m,n) = m + n + 1$ is **ONTO**

Proof: For any $k \in \mathbb{Z}$, choose $m = k - 1$ and $n = 0$. Then $f(k-1, 0) = (k-1) + 0 + 1 = k$. $\therefore$ Onto.

(e) $f(m,n) = m^2 - 4$ is **NOT ONTO**

Counterexample: Consider $k = 1$. We need $m^2 - 4 = 1$, so $m^2 = 5$. Since $\sqrt{5}$ is irrational, no integer m satisfies this. $\therefore$ Not onto.

Question 7: Examples of Functions from $\mathbb{N}$ to $\mathbb{N}$

Category	Example Function	Why?
(a) One-to-one, not onto	$f(n) = n + 1$	Injective but misses 0 (or 1)
(b) Onto, not one-to-one	$f(n) = \lfloor n/2 \rfloor$	Surjective but $f(2)=f(3)=1$
(c) Bijection (both)	$f(n) = n$	Identity is bijective
(d) Neither	$f(n) = 1$	Constant function

Detailed Explanations:

(a) One-to-one but NOT onto: $f(n) = n + 1$

One-to-one: If $f(n_1) = f(n_2)$, then $n_1 + 1 = n_2 + 1$, so $n_1 = n_2$. ✓

Not onto: The value 0 (or 1 if $\mathbb{N}$ starts at 1) is never in the range, since $n + 1 \geq 1$ (or ≥ 2) for all $n \in \mathbb{N}$. ✗

(b) Onto but NOT one-to-one: $f(n) = \lfloor n/2 \rfloor$

Onto: Every $k \in \mathbb{N}$ can be achieved by $f(2k) = k$. ✓

Not one-to-one: $f(2) = 1$ and $f(3) = 1$, but $2 \neq 3$. ✗

(c) Bijection (one-to-one AND onto): $f(n) = n$

Identity function: Obviously both one-to-one and onto. ✓✓

(d) Neither: $f(n) = 1$

Constant function: $f(0) = f(1) = 1$ (not one-to-one), and only 1 is in range (not onto). ✗✗

Question 8: Bijections from $\mathbb{R}$ to $\mathbb{R}$

Part	Function	Bijection?	Reason
(a)	$f(x) = -3x + 4$	YES ✓	Linear, invertible
(b)	$f(x) = -3x^2 + 7$	NO ✗	Parabola, not one-to-one
(c)	$f(x) = (x+1)/(x+2)$	NO ✗	Range excludes 1
(d)	$f(x) = x^5 + 1$	YES ✓	Strictly increasing, onto

Detailed Analysis:

(a) $f(x) = -3x + 4$ is a **BIJECTION**

One-to-one: If $-3x_1 + 4 = -3x_2 + 4$, then $x_1 = x_2$. ✓

Onto: For any $y \in \mathbb{R}$, solve $y = -3x + 4$ to get $x = (4-y)/3 \in \mathbb{R}$. ✓

Inverse: $f^{-1}(x) = (4-x)/3$

(b) $f(x) = -3x^2 + 7$ is **NOT** a bijection

Not one-to-one: $f(1) = 4$ and $f(-1) = 4$, but $1 \neq -1$. ✗

Not onto: Range is $(-\infty, 7]$, cannot achieve $y = 8$. ✗

(c) $f(x) = (x+1)/(x+2)$ is **NOT** a bijection

Not onto: For $y = 1$: $(x+1)/(x+2) = 1$ implies $x+1 = x+2$, which is impossible. Range is $\mathbb{R} \setminus \{1\}$. ✗

(d) $f(x) = x^5 + 1$ is a **BIJECTION**

One-to-one: x^5 is strictly increasing (monotonic), so $x^5 + 1$ is too. ✓

Onto: For any $y \in \mathbb{R}$, $x = \sqrt[5]{y-1} \in \mathbb{R}$ (fifth root defined for all reals). ✓

Inverse: $f^{-1}(x) = \sqrt[5]{x-1}$

Question 9: Graph of $f(n) = 1 - n^2$ from $\mathbb{Z}$ to $\mathbb{Z}$

Since the domain is $\mathbb{Z}$ (integers), the graph consists of discrete points, not a continuous curve.

Graph of $f(n) = 1 - n^2$

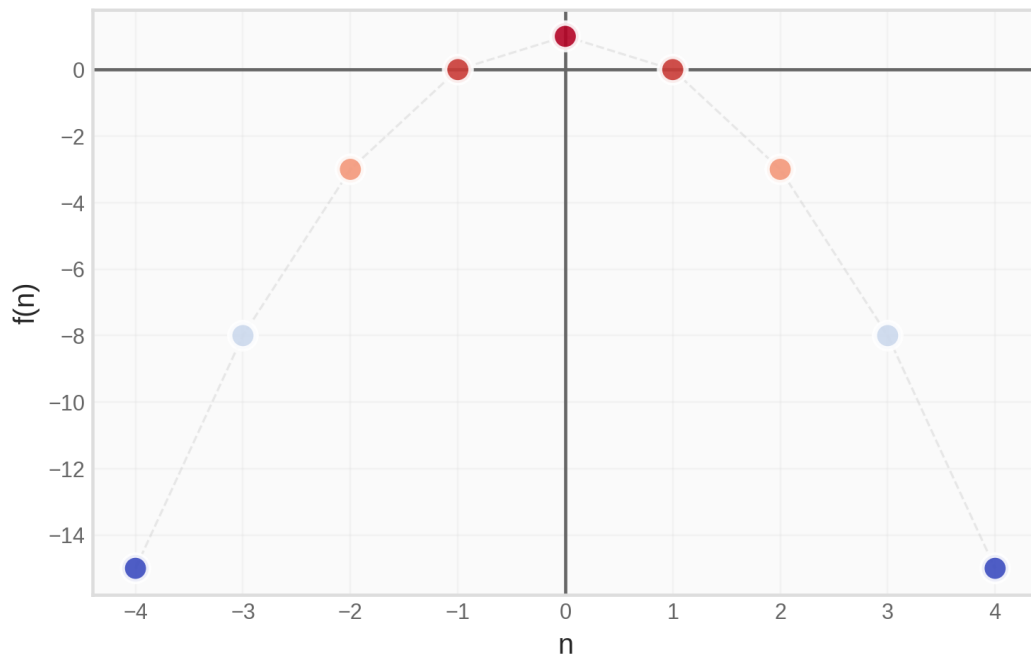


Table of values:

n	$f(n) = 1 - n^2$
-4	-15
-3	-8
-2	-3
-1	0
0	1
1	0
2	-3
3	-8
4	-15

Key observations:

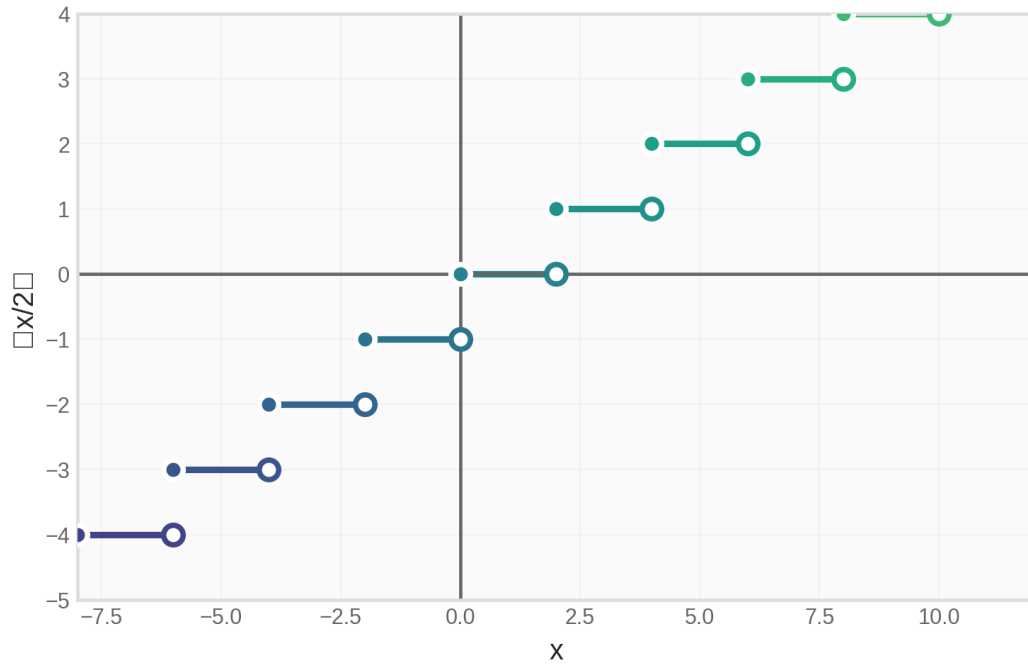
- The graph is symmetric about the y-axis (even function)
- Maximum value is 1 (at $n = 0$)

- Points form a downward-opening parabola shape
- Graph consists of discrete points only (domain is integers)

Question 10: Graph of $f(x) = \lfloor x/2 \rfloor$ from $\mathbb{R}$ to $\mathbb{R}$

This is a step function where the value stays constant over intervals and jumps at even integers.

Graph of $f(x) = \lfloor x/2 \rfloor$



Function behavior:

Interval	$f(x)$ value
$[-4, -2)$	-2
$[-2, 0)$	-1
$[0, 2)$	0
$[2, 4)$	1
$[4, 6)$	2
$[6, 8)$	3
$[8, 10)$	4

Key features:

- Horizontal segments at integer heights
- Jumps occur at even integers: $x = \dots, -4, -2, 0, 2, 4, 6, \dots$
- Left endpoints are included (closed circles)
- Right endpoints are excluded (open circles)

Question 11: Inverse of $f(x) = x^3 + 1$

To find the inverse, we solve for x in terms of y :

Step	Equation
1. Original function	$y = x^3 + 1$
2. Isolate x^3	$y - 1 = x^3$
3. Take cube root	$x = \sqrt[3]{y - 1}$

Answer: $f^{-1}(x) = \sqrt[3]{x - 1}$

Verification:

Composition	Result
$f(f^{-1}(x))$	$(\sqrt[3]{x-1})^3 + 1 = (x-1) + 1 = x \checkmark$
$f^{-1}(f(x))$	$\sqrt[3]{(x^3+1) - 1} = \sqrt[3]{x^3} = x \checkmark$

Alternative notation: $f^{-1}(x) = (x - 1)^{1/3}$

Note: The cube root function is defined for all real numbers (unlike square root), so the inverse function has domain $\mathbb{R}$, matching the range of the original function.